Sage – Appointments App

AI Product Intern Assignment • Documentation & Technical Notes

Kayro Submission

WHAT WE WERE ASKED

Build a small, high-fidelity appointments product with two sides sharing the same data: a mobile-first **Client App** (home screen with next visit, booking with zero double-booking, upcoming/past lists, reschedule/cancel, consumer wellness feel) and a **Nutritionist Dashboard** (appointment list with name, phone, date, time, and colored statuses: Booked, Done, No-show; manual entry; Today filter; two-way sync; loaded with sample data). The solution must also include at least one mechanism mitigating the 25% client no-show rate.

YOUR APPROACH

I built an integrated single-page application containing two views (Client App and Nutritionist Dashboard) connected to a unified in-memory and local persistence layer. Both sides read and write the exact same appointments store with instant reactive updates across views and browser tabs. The client experience prioritizes calm, reassuring wellness aesthetics (sage teal `#1f6f6b`, soft peach badges, Figtree typography, dark mode support) centered in a mobile viewport, while the dashboard provides a high-density, actionable clinical interface. To tackle no-shows, I embedded three complementary mechanisms: an active client confirmation button, a 1-tap WhatsApp reminder link, and a real-time no-show rate KPI.

DECISIONS AND ASSUMPTIONS

- **Sample Data Normalization:** The sample data arrived with 3 date formats (`DD/MM/YYYY`, `YYYY-MM-DD`, `DD-MM-YYYY`) and 4 time notations (`10:00`, `11:30am`, `4 pm`, `14:00`). I implemented standardizing parsers converting dates to ISO (`YYYY-MM-DD`) and times to 24h (`HH:mm`). `24/10/2026` was parsed as day/month since month 24 does not exist. Phone numbers are normalized to 10 digits and formatted as `98200 11111`. Riya Shah's two entries share the same phone and are correctly recognized as one client with two separate visits. Meera Iyer has no phone and is labeled "No phone" without generating fictitious numbers.
- **Shared Data & Two-Way Sync:** Data lives in `localStorage` with a `storage` event listener. Any appointment booked or rescheduled on the client side updates the dashboard immediately (even in another browser tab), and any status change or manual booking made by the nutritionist reflects instantly in the client view.
- **Demo Date Anchor vs. Today Filter:** Because the assignment's sample records are set in October 2026, viewing the app with a real device clock would normally cause the home screen to show "No visit planned" and the Today filter to show 0 appointments. I implemented a smart **Demo Date Anchor (24 Oct 2026)** so evaluators immediately experience the active appointments, with a 1-click toggle to switch to their live device clock.
- **No Login / Profile Switching:** Rather than artificial authentication, clients are identified by phone number (defaulting to Riya Shah). Quick-switch chips allow evaluators to alternate between Riya, Arjun, and Kabir in one click.
- **Time Slots & Double-Booking:** Consultations run every 30 minutes (9:00 AM – 5:30 PM). Slots are blocked if already booked or in the past, validated both at selection time and upon submission.
- **Cancellation:** Following the brief's three official statuses, cancellation removes the reservation to instantly release the slot.

WHAT YOU ADDED

Beyond the brief, I added: (1) **"I'll be there" confirmation button** on the client home screen that broadcasts a **"✓ Client confirmed"** tag to the nutritionist; (2) **One-tap "Remind on WhatsApp"** pre-filled messaging link (`wa.me`); (3) **Clinical No-show Rate metric card** (`No-shows / [Done + No-shows]`) measuring practice attendance; (4) **1-Click Client Profile Switcher** for fast multi-user testing; (5) **"Past: please update"** alert tags for overdue booked appointments; and (6) **"↺ Reset Sample Data"** button to restore the clean assignment state at any time.

WHAT THE AI GOT WRONG (AND HOW YOU CAUGHT AND FIXED IT)

- **Missing Phone Validation on Rescheduling:** The AI initially used a global phone validation rule on reschedule. For clients who booked without a phone number (such as Meera Iyer), rescheduling threw an un-fixable "*Enter a valid 10-digit phone number*" error because the phone input field is intentionally hidden during reschedule. I caught this during boundary-case testing and updated the logic to preserve existing phone-less records.
- **Evaluator Date Mismatch (Empty Initial State):** The AI initially coded relative comparisons directly against `new Date()`. Because the sample dataset is dated October 2026, evaluating the app on any other date rendered Riya's consultations as "past", leaving the client hero on "No visit planned" and the Today filter empty. I caught this while testing first-load impressions and introduced the Demo Date controller.
- **Cross-Tab Reactive Sync:** The initial AI code only mutated local component state without triggering cross-tab sync. I added the `window.storage` subscriber and single-source mutation helper to guarantee instant two-way synchronization.

HOW TO RUN IT

Open `index.html` directly in any modern browser—no node modules, build tools, or server setup required. For cloud deployment, push to GitHub and link to Vercel using the "Other" (static) framework preset.